SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17** 03

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

Duration : 45 Minutes
QUESTIONS: 2 **Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:

I D.VAISHNAVI (192524143) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D.Vaishnavi

18/20

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
   Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

CO1 AT-3
[DESCRIPTIVE TEST]

COURSE CODE: CSA1703

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COURSE: ARTIFICIAL INTELLIGENCE

1. Smart Home Automation system:

• A smart home Automation system is an intelligent system that automatically controls lighting, Temperature, and security by using the sensors and user preferences.

• It Receives the information from the Environment process it, and performs the Actions to achieve the Goals.

Types of Agents:

→ Simple Reflex Agent

→ Model-based Agent

→ Goal-Based Agent

→ Utility-Based Agent

Simple-Reflex Agent:

Simple Reflex Agents, it works only on the current input and follows the condition-action Rules.

Ex: If Motion is detected → Turn ON Lights

If No motion is detected → Turn OFF Lights

Advantages: • Fast and Easy to implement
• Suitable for simple tasks and Actions
To perform the Goals.

Disadvantages: • It cannot remember the previous Events.
• It cannot Learn the user preferences.

Model- Based Agent:
• Model- based Agent keeps an internal model of Environment and Remember the previous States.

Ex: If the user|owner leaves the houses, the System switch off lights, lock the doors, and activate the security Automatically.

Advantages:
• uses memory for better decisions.
• works well in changing the Environments

Disadvantages: More complex than the simple Reflex Agent.

Goal- Based Agents:
It selects an actions to Achieve a specific Goals.

Ex: Goal: Maintain Room Temperature at 24°c

• If Temperature increases, switch ON the air conditioner

• If Temperature decreases, switch OFF the AC.

Advantages:

→ Making an intelligent decision-making

→ Helps in to Achieve the desired Goals.

Disadvantages:

→ Requires planning, so it takes time to reach Goals.

Utility-Based Agent:

• It chooses action that gives the high and overall benefit for multiple factors.

Ex: It balances : user comfort, Energy Efficiency, home security.

Advantages: → produces the overall performance

→ optimizes the Electricity, Energy consumption.

Disadvantages:

→ It is more complex and Expensive. ~~for~~ consumption, utility Based Agent in overall action performance. of smart home

Justification:

These are the Most Effective for a Smart Home Automation System.

Reasons:
- Learns user habits
- Achieve the goals such as comfort, security
- Minimizes the Electricity consumption
- Makes an intelligent decisions in real-time

Conclusions:

The Intelligent Agents provides the better comfort, improved security, and Energy Efficiency, these are the best choice for Smart home Automation system.

2. Robot Navigation in Uniform Maze search:

- A Robot is an AI placed Environment, it learns unknown maze must search for the Exist Algorithms help the robot Explore possible pathways and Reach the Destination.

Uniform cost search [UCS]:

→ UCS Expands the node with the Lowest cost path.

Working:

→ It starts from the initial position

↓

Explores paths with minimum total cost path.
↓
It Stops when it Reach Destination

Advantages:-

- Finds the optimal path solution
- It works with when path costs are different

Disadvantages:-

- High memory usage
- Slow for large search mazes

Complexity:-

Time : $O(b^d)$ $b \rightarrow$ branching factor

Space: $O(bd)$ $d \rightarrow$ Depth solution

ii) IDS: [Iterative Deepening search]

- IDS combines both Depth first search and Breadth First Search Algorithm

Working: • search depth 0
↓
• and then, move to depth 1, depth 2,
↓
• continues until it reaches the Goal and Exits.

Advantages:

- Low memory usage
- complete path solution
- Find the shortest route

Disadvantages:

- Repeats node Expansions
- Slower due to Repeated search

Complexity:

Time: $O(b^\wedge(1+C^* / E))$

Space: $O(b^\wedge(1+C^* / E))$

b → branching factor

C → optimal path

E → minimum cost path

Justification:

- Goal-Based Agent focuses on reaching the destination
- UCS provides optimal path solution
- IDS is memory usage Efficient for large unknown maze search.

Conclusion:

UCS is best solution for finding the optimal path is an unknown maze search. If memory is limited and all moves have Equal cost, IDS is the best choice for Goal based Agent, Robot Navigation System.